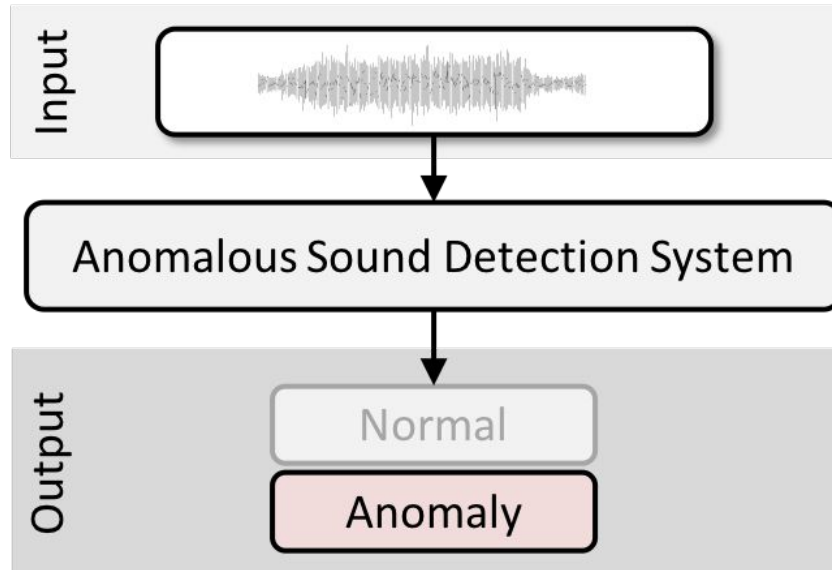


Detección de Anomalías Acústicas en Maquinaria Industrial

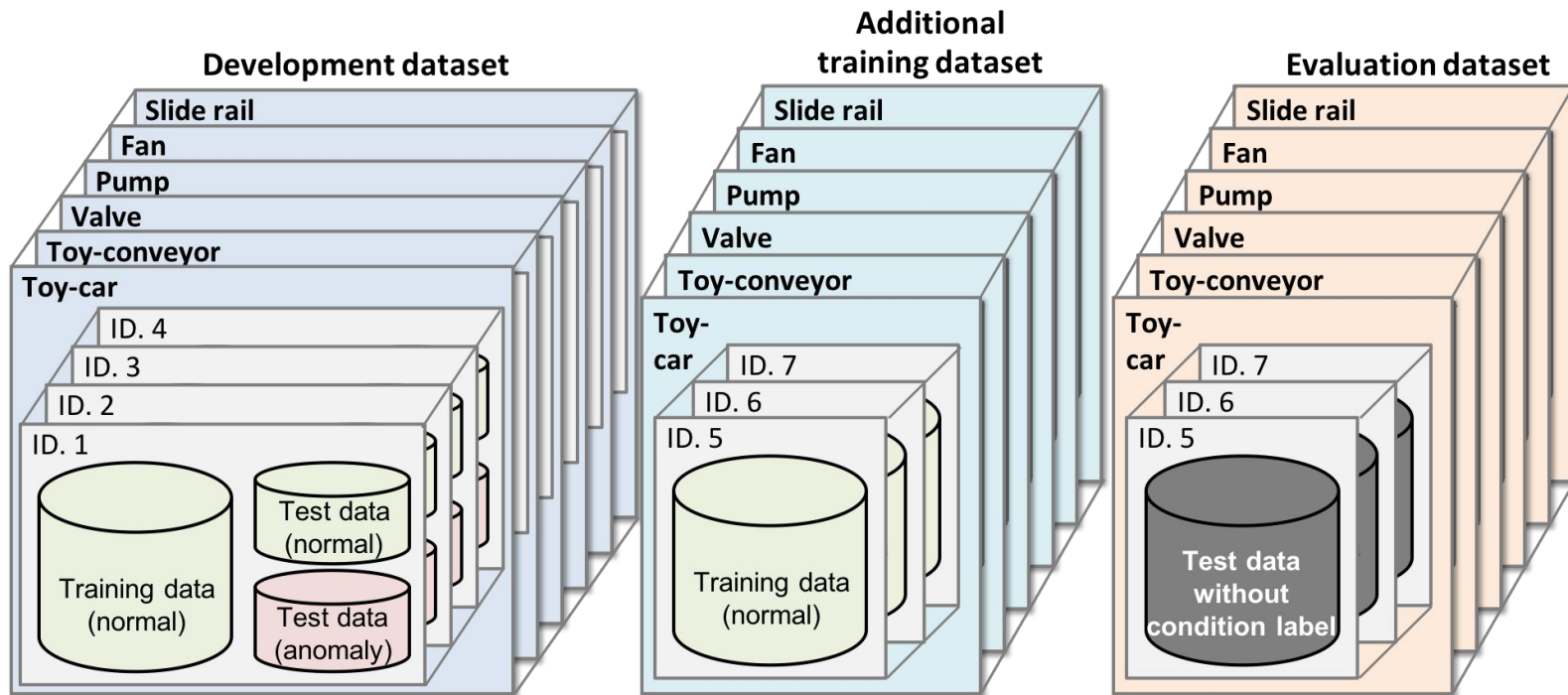
Arley Andrés Villamizar Molina

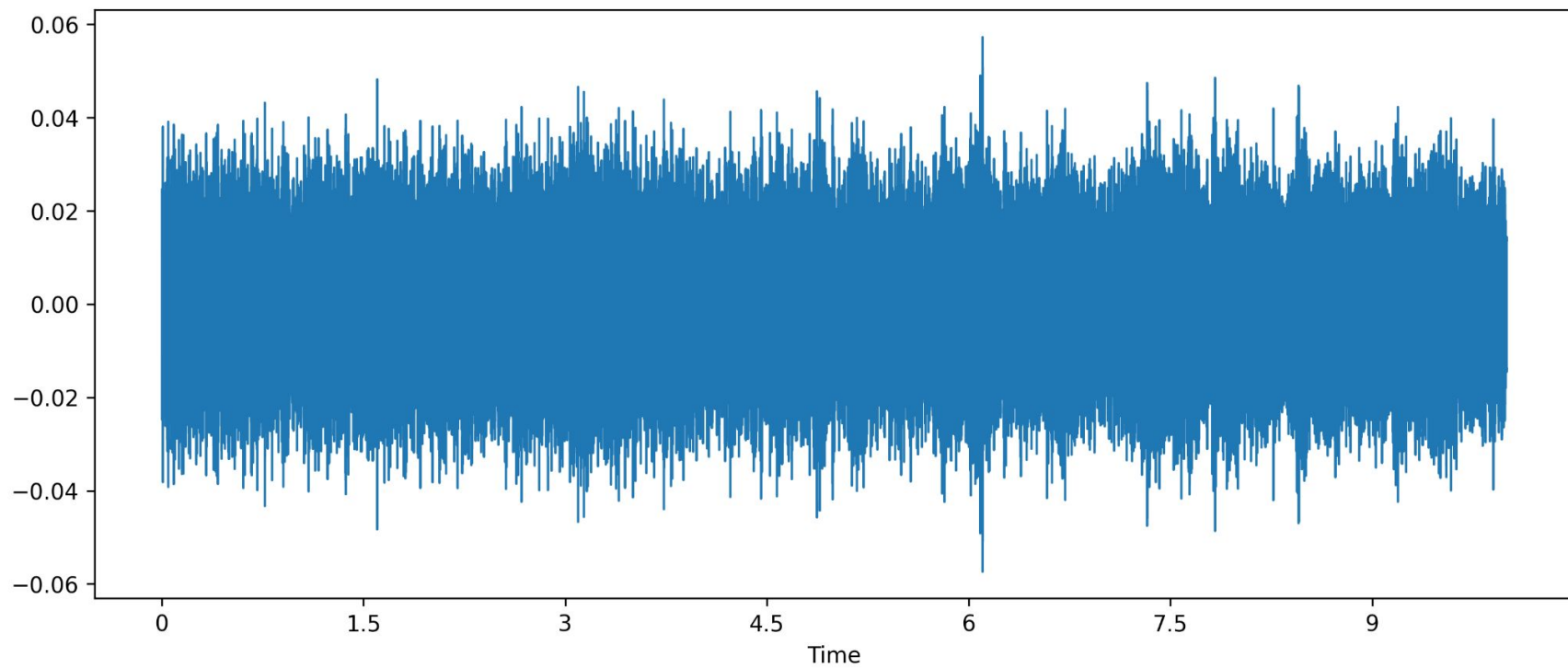
Alcance

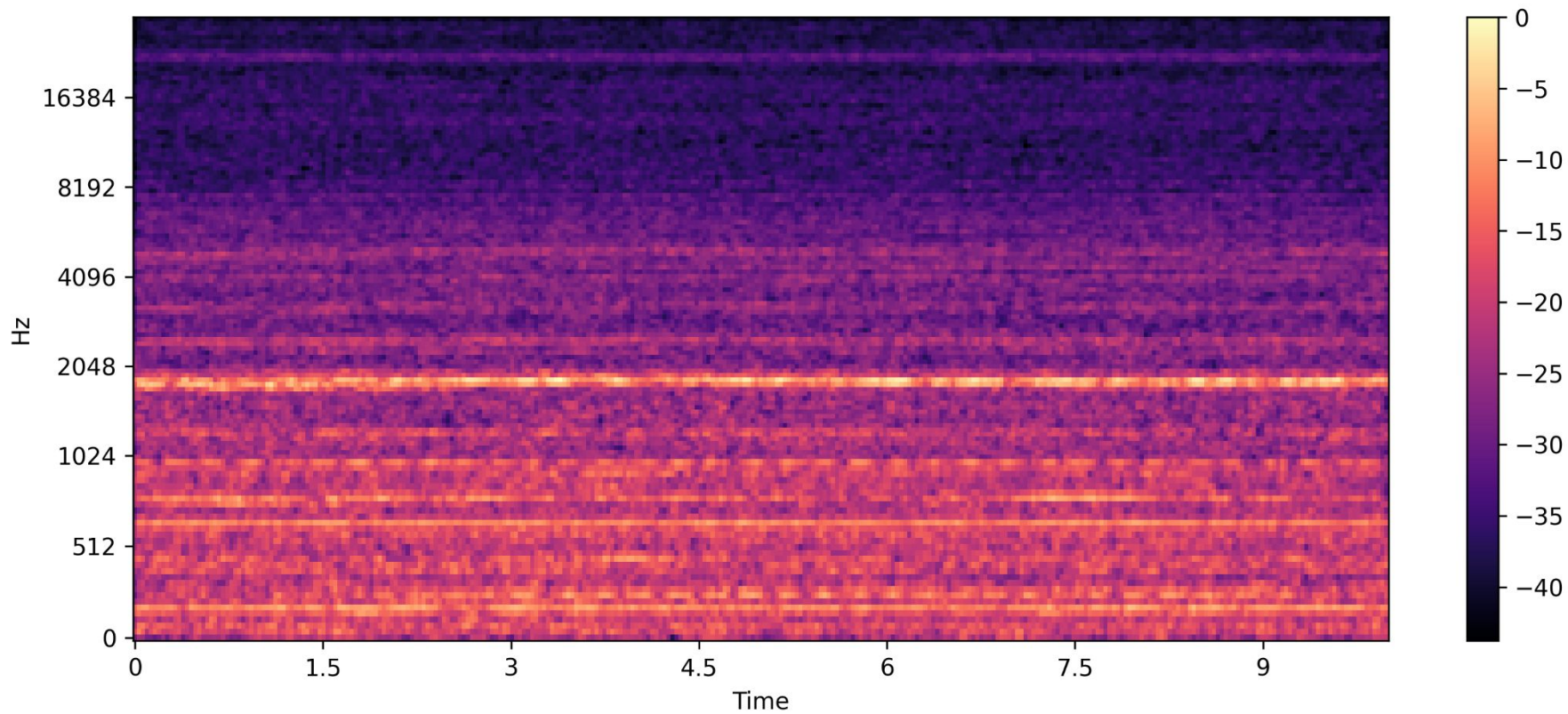


Detectar anomalías
acústicas en maquinaria
industrial haciendo uso
de autoencoders

Dataset



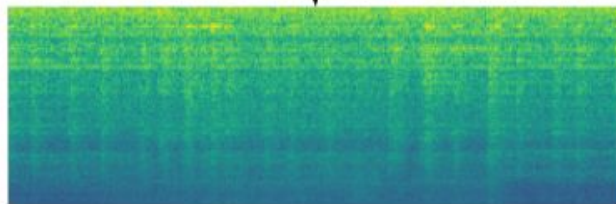




Audio Signal



Log-mel energies spectrogram

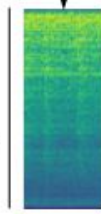


Buffer of 1 second

0.9 seconds overlap

Buffer of 1 second

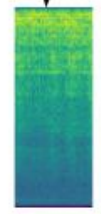
128 MFECs



32 frames

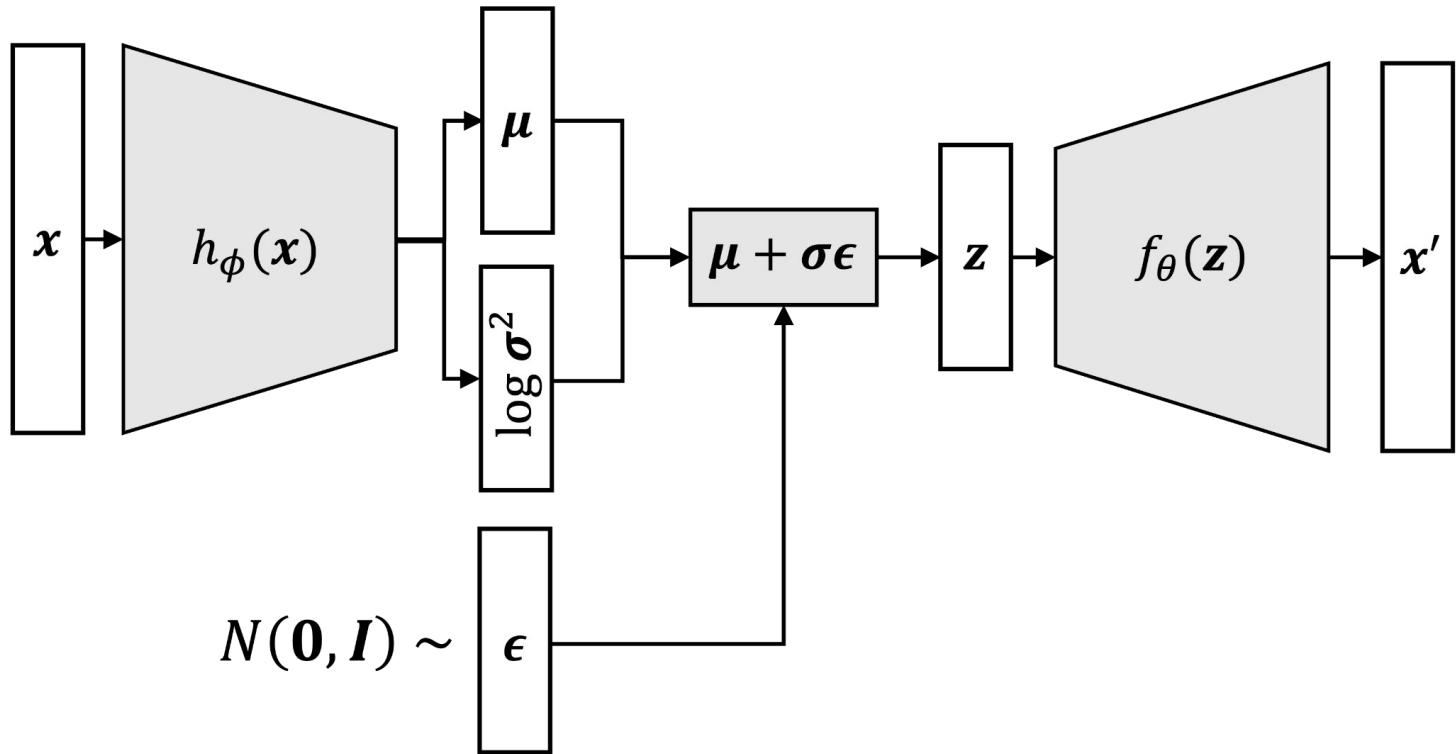
N data segments

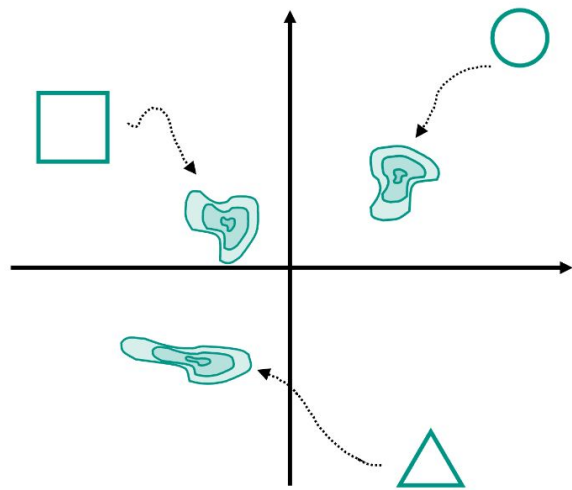
128 MFECs



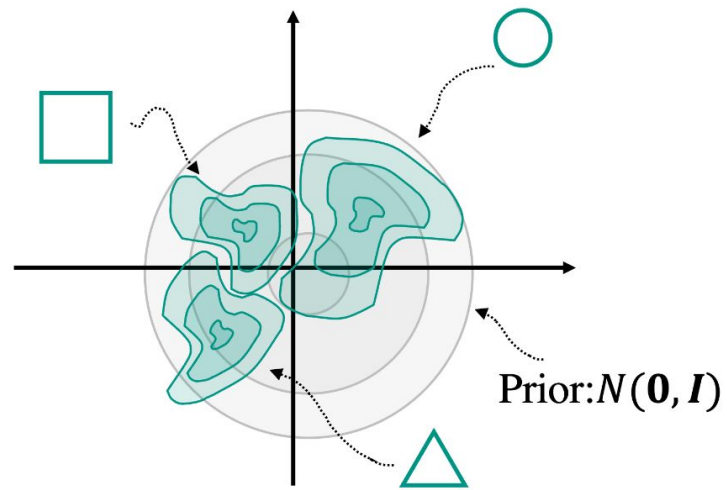
32 frames

Variational Autoencoder

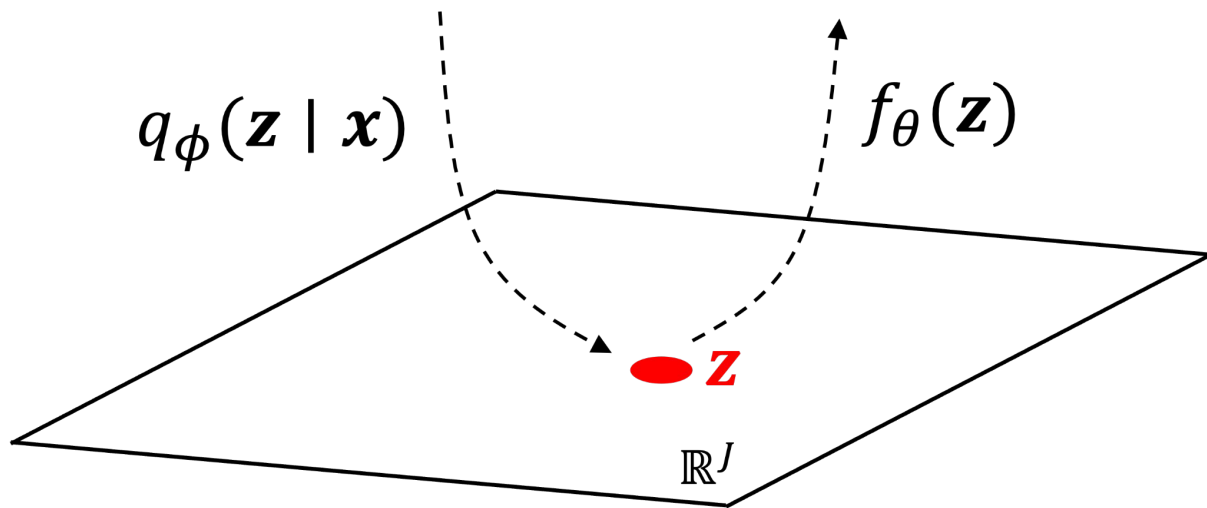
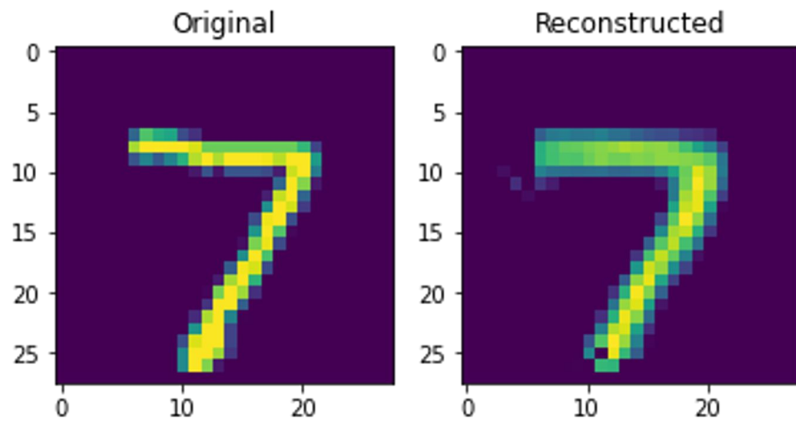


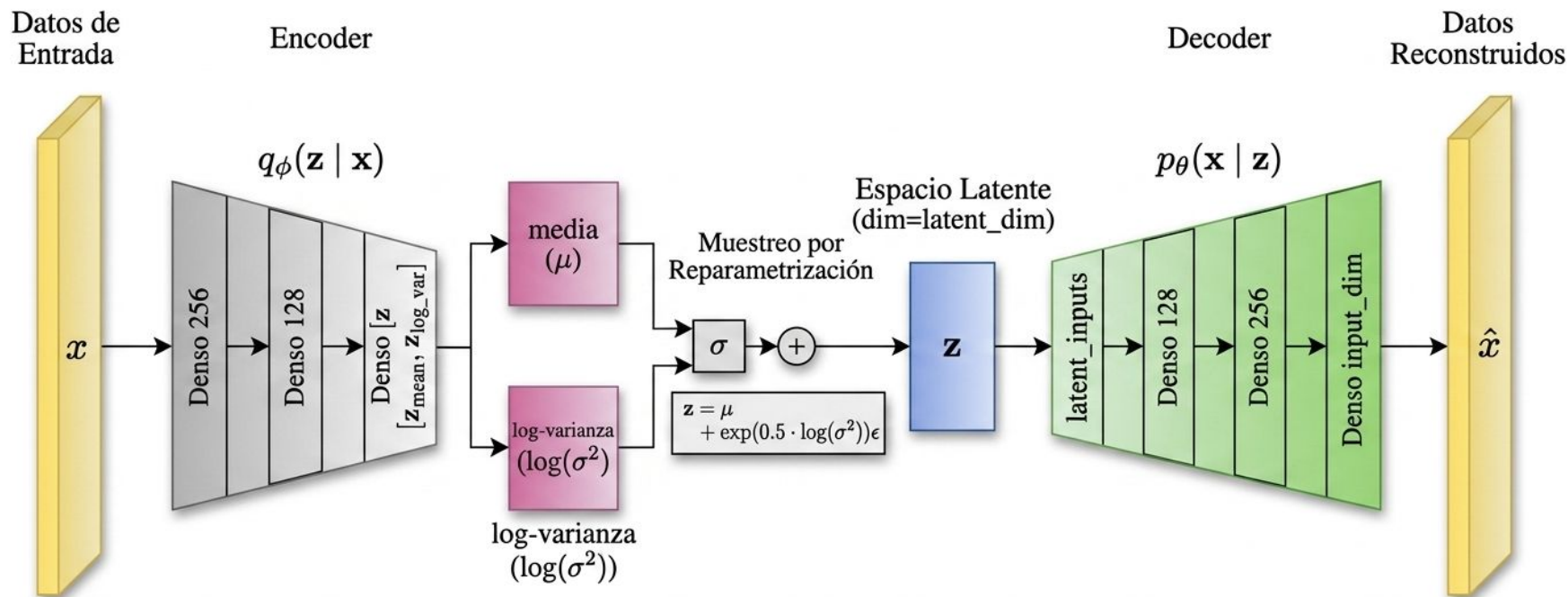


AE latent space



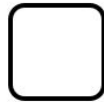
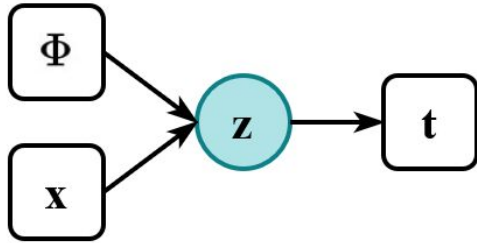
VAE latent space





$$\text{Pérdida Total} = \mathbb{E}_{\mathbf{z} \sim q(\mathbf{z} | \mathbf{x})} [\log p(\mathbf{x} | \mathbf{z})] - 0.5 \cdot \sum (1 + \log(\sigma^2) - \mu^2 - \exp(\log(\sigma^2))) + \text{Divergencia KL}$$

Original Formulation

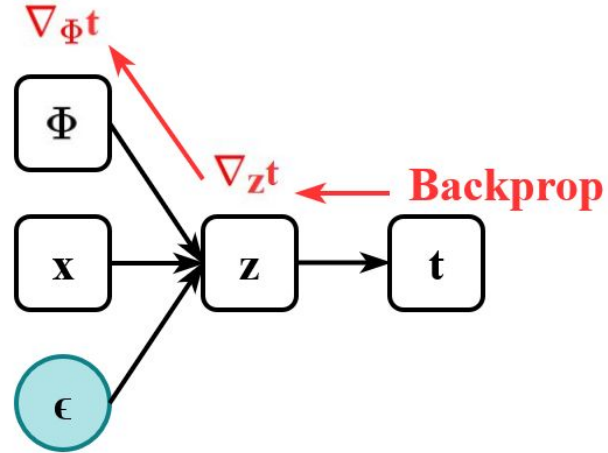


Deterministic
Node



Random
Node

Reparameterized Formulation



Feedforward

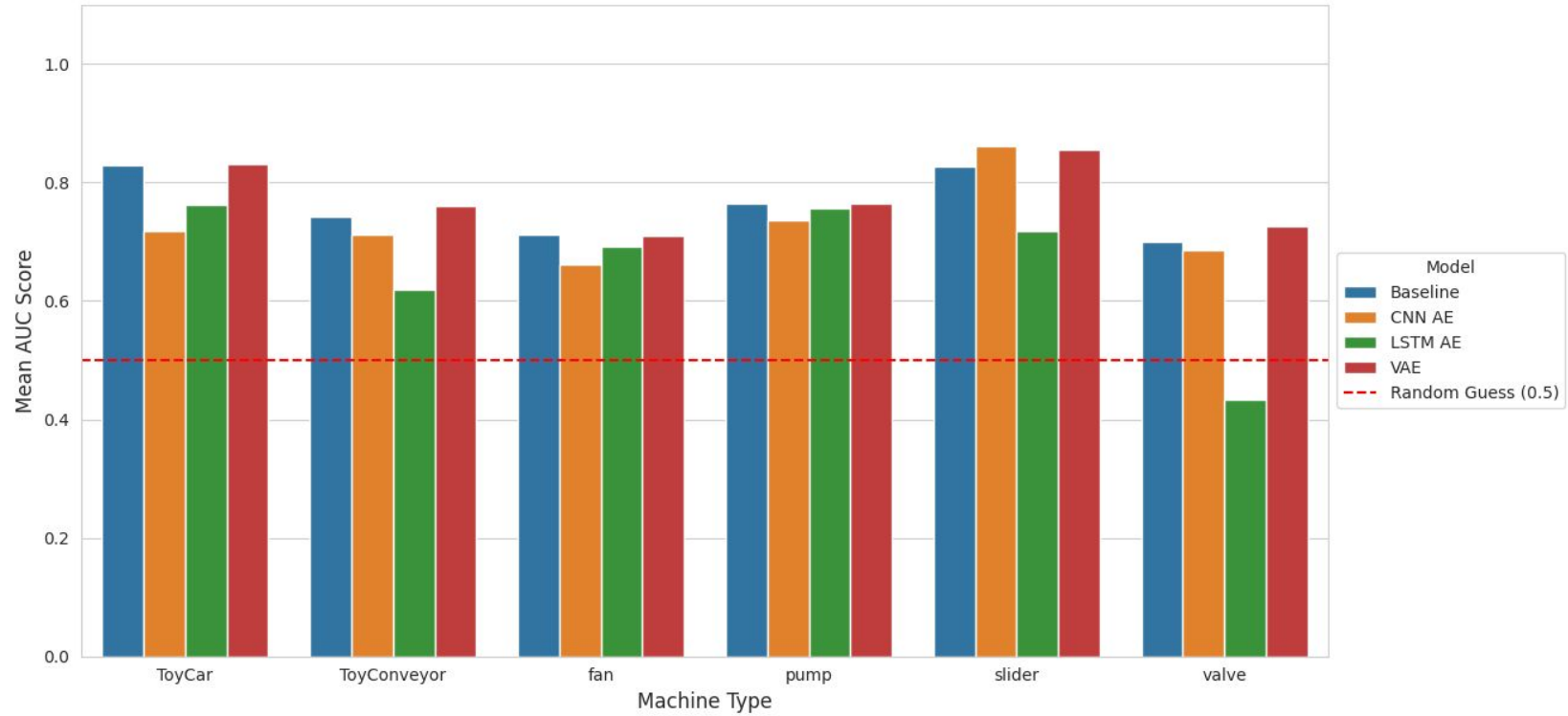


Differentiation

$$\text{loss}_{\text{VAE}}(\phi, \theta) = - \sum_{i=1}^n E_{\mathbf{z}_i \sim q_{\phi}(\mathbf{z}_i | \mathbf{x}_i)} [\log p_{\theta}(\mathbf{x}_i | \mathbf{z}_i)] - KL(q_{\phi}(\mathbf{z}_i | \mathbf{x}_i) || p(\mathbf{z}_i))$$

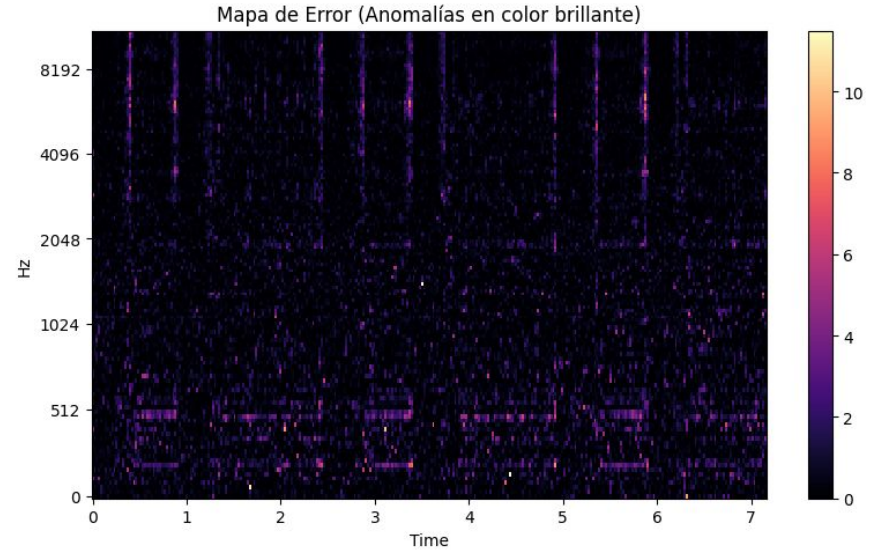
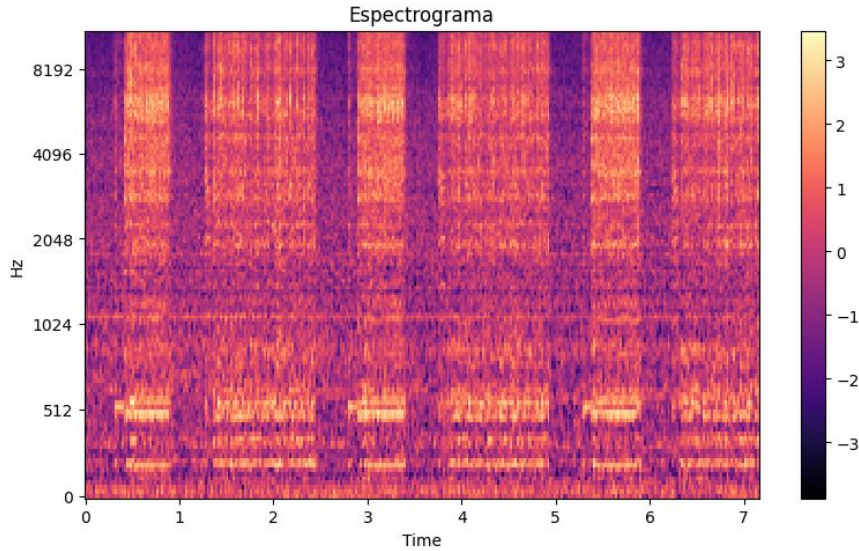
$$KL(q_{\phi}(\mathbf{z} | \mathbf{x}) || p(\mathbf{z})) = -\frac{1}{2} \sum_{j=1}^J \left(1 + \log \sigma_j^2 - \mu_j^2 - \sigma_j^2 \right)$$

Comparación de Modelos por Tipo de Máquina (Mean AUC)

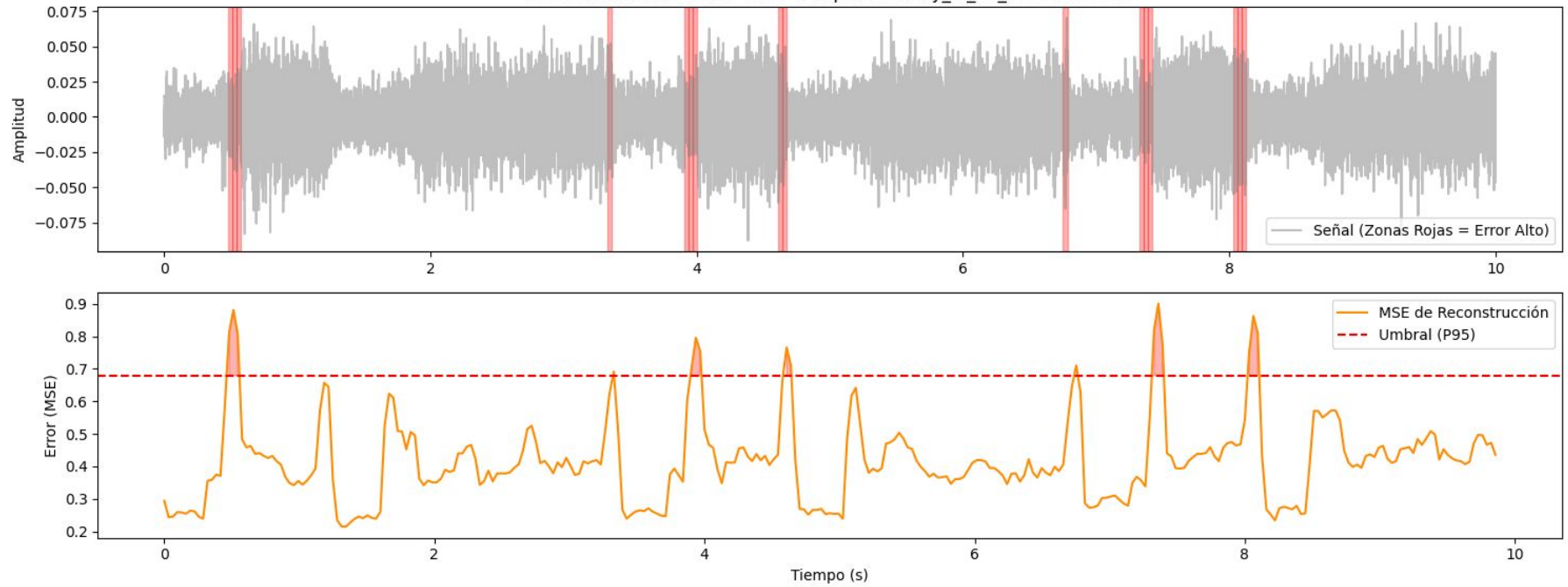


Modelo	Promedio AUC
AE	0.763728
CNN AE	0.730149
LSTM AE	0.666109
VAE	0.775723

Mapa de Errores



Detección de Anomalías en el Tiempo: anomaly_id_00_00000000.wav



Notebook